A Double Dictionary-Based Nonlinear Representation Model for Hyperspectral Subpixel Target Detection

Xiaoyi Wang, Liguo Wang, Hao Wu, Jiawen Wang, Kaipeng Sun, Anqi Lin, and Qunming Wang

***Abstract*— Due to the limitations of hardware technology and budget constraints, there always exists a tradeoff between spatial and spectral resolutions in a hyperspectral image (HSI). Because of the limited spatial resolution, mixed pixels are a common issue in HSIs, and consequently, some targets appear as subpixels. The effectiveness of hyperspectral target detection is affected greatly by the subpixel targets, especially when the size of the targets is small. In this article, we proposed a double dictionary-based nonlinear representation model for hyperspectral subpixel target detection (DDNRTD). DDNRTD represents HSIs with a nonlinear model based on background and target dictionaries, which fully considers the spatial property of background and targets and can separate background and targets reliably, especially for small-sized subpixel targets. In addition, we designed an over-completed background dictionary construction strategy to represent the background part more effectively, which integrates spectral angle distance (SAD) with sparse representation. Experiments on two simulated and five real HSI datasets showed that the proposed DDNRTD method produced more accurate detection results than six state-of-the-art methods.**

***Index Terms*— Double dictionaries, hyperspectral target detection, nonlinear representation, subpixel target.**

I. Introduction

HYPERSPECTRAL target detection techniques have been widely used in military reconnaissance, geological exploration, ocean monitoring, and precision agriculture [1]–[4]. In theory, target detection is a binary classification problem: according to the spectral difference, the pixels in hyperspectral image (HSI) are divided into target or background. However, unlike common image classification tasks, target detection aims to detect targets with small volume and low occurrence probability in HSIs. Therefore, the rules for common classification (e.g., classification error minimization based on a number of samples) are not suitable for target detection.

Over the past decade, a variety of hyperspectral target detection methods have emerged. As different materials present different spectral characteristics in HSIs, target detection can be realized by comparing the spectral similarity between the tested pixel and known target spectrum. The most common measurement methods are based on Euclidean distance (ED) and spectral angle mapping (SAM) [5]. Although the target detection methods based on ED and SAM have the advantages of simple form and short computational time, their detection accuracy may not meet the requirement for practical applications due to the great spectral variation. Alternatively, there are a number of target detection methods based on statistics. For example, spectral matched filtering (SMF) [6] works under the assumption that the target and the background pixels are normally distributed with the same covariance matrix but different mean values. The constrained energy minimization (CEM) method [7] designs a finite impulse filter using all pixels of an HSI. Then, the HSI is input into the finite impulse filter to constrain the output energy of the background and highlight the response of the target. On the basis of CEM, Zou and Shi [8] proposed a hierarchical suppression method that uses a layer-by-layer filtering procedure, named hierarchical CEM (hCEM). Chang [9] proposed the orthogonal subspace projection (OSP) algorithm, which underpins the development of signature space orthogonal projection [10], target signature space orthogonal projection [11], and oblique subspace projection.

In recent years, a number of novel hyperspectral target detection methods have been developed. Zhang *et al.* [12] combined sparse representation with binary hypothesis theories and proposed a sparse representation-based binary hypothesis (SRBBH) model for hyperspectral target detection. Bitar *et al.* [13] introduced the low-rank and sparse matrix decomposition (LRSMD) theory into hyperspectral target detection, leading to sparse and low-rank matrix decomposition (SLRMD) for automatic target detection in hyperspectral imagery. It works on the principle that the background part has low-rank property and the target part has global sparsity. The target part can be represented by the target

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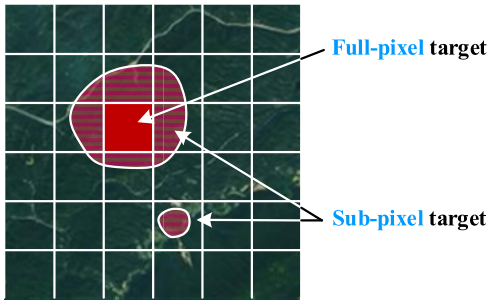


Fig. 1. Full-pixel and subpixel targets in HSI.

dictionary and the corresponding representation coefficient matrix. Low-rank and sparse decomposition theory has also been applied to hyperspectral anomaly detection. Qu *et al.* [14], [15] combined spectral unmixing with dictionary-based low-rank and sparse decomposition method and proposed a hyperspectral anomaly detection method using abundance vector instead of the original HSI. More recently, deep learning has attracted extensive attention in the field of remote-sensing image processing [16]–[19]. Shi *et al.* [20] combined few-shot learning and domain adaptation and introduced modulated deformable convolution (MDConv) operation and a discriminatively boosted loss function into the learning model for target detection, which overcomes the limitations of insufficient training samples and sensor-dependent transferability to some extent. Zhu *et al.* [21] combined the target prior spectrum and typical background samples obtained by hybrid sparse representation and classification-based pixel selection strategy, to construct sufficient target samples. Then, a two-stream convolutional neural network was used to extract the spectral features of the background and the target.

Due to the limited spatial resolution, there are usually a large number of mixed pixels in HSIs. The problem of mixed pixels has been attracting increasing attention in many HSI processing issues, such as spectral unmixing [22], [23], super-resolution mapping [24]–[26], and HSI classification [27], [28]. For hyperspectral target detection, if the target covers the entire pixel, it is called a full-pixel target, as shown in Fig. 1. The full-pixel target can be considered as a strong target because of relatively obvious spectral differences between them and background. Normally, the existing target detection methods can detect most full-pixel targets with acceptable accuracy. When the target covers only part of the pixel (i.e., the pixel contains both background and target components), it is called a subpixel target (see Fig. 1). In this case, the spectrum of the observed pixel (i.e., mixed pixel) is a combination of the spectrum of the background and the target, and the weights can be viewed as the proportions of area covered by the background and the target. For the subpixel target, when it occupies a small area in the pixel, the spectral difference between the observed pixel and background will be small, and such a pixel is called as a weak target.

In terms of unsupervised subpixel target detection (i.e., anomaly detection), there exist several studies. For example, Duran and Petrou [29] proposed a subpixel anomaly detection method based on unconstrained spectral unmixing and the use of the negative and superunity abundances. Khazai *et al.* [30] identified the most appropriate feature for discriminating the target and the local background pixels. Huang *et al.* [31] utilized spectral unmixing, morphological operation, and super-pixel segmentation methods to extract subpixel, pixel, and super-pixel features from HSIs, and the multiscale features were integrated into a final target detection map by decision fusion.

Up to now, some researches on subpixel target detection have been carried out. The adaptive coherence estimator (ACE) method [32] works on the assumption that when the targets are at subpixel resolution, the background covariance matrix needs to be scaled. Manolakis *et al.* [33] proposed the adaptive matched subspace detector (AMSD) based on a linear mixed model, assuming that each background pixel can be represented by the product of background endmembers and abundance matrix. Broadwater and Chellappa [34] integrated the fully constrained least-squares algorithm with AMSD and ACE and proposed two hybrid detectors for the subpixel target. Generally, ACE estimates the background covariance matrix using all pixels. Thus, when there are many target pixels in the HSI, the detection accuracy will be affected greatly. AMSD uses the linear mixed model to characterize the profile of background and target, but most HSI do not meet this assumption. Inspired by sparse representation theory, Chen *et al.* [35] proposed a sparsity-based target detection algorithm (STD). STD works on the principle that the background and target pixels can be sparsely represented by the trained background and target dictionaries, respectively, in which the background dictionary atoms are selected by a sliding dual-window. However, STD could not fully consider the spatial characteristics of background and target.

To address the above issues, in this article, we proposed a double dictionary-based nonlinear representation model for hyperspectral subpixel target detection (DDNRTD). Based on the assumption that the target has global sparsity and background has low-rank property in HSIs, the reweighted nuclear norm and $\ell_{2,1}-$norm as low-rank and sparsity constraints are imposed on the background and target representation coefficient matrices, respectively. Compared with standard nuclear norm, the reweighted nuclear norm can enhance the low-rank property of the matrix through global iterative optimization, and meanwhile, the $\ell_{2,1}-$ norm can realize the structural sparsity of the matrix. There are three main contributions of the proposed method.

1) The scheme of integrating LRSMD with double dictionaries for nonlinear representation of HSIs not only extracts the subpixel target components in mixed pixels, but also effectively separates the background and target parts and reduces the false alarm rate.
2) Since the target spectrum is assumed to be known in the model, the method to construct the background directly affects the representability of the background. In this article, we propose a spectral angle distance (SAD) and sparse representation background

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dictionary (SSBD) method for more reliable background dictionary construction.

3) A popular method is used for solving the convex optimization problems in DDNRTD with separable structures, namely alternating direction method of multipliers (ADMM) [36]. It has more satisfactory detection performance than some state-of-the-art hyperspectral target detection methods, especially for subpixel targets.

The rest of this article is organized into four sections. The proposed DDNRTD method including the background dictionary construction strategy is presented in Section II. Section III presents the experimental results based on two simulated and five real HSI datasets, and these experimental results are further discussed and analyzed in Section IV. Finally, the conclusion is presented in Section V.

II. Methods

In this article, the HSI is processed based on 2-D matrix decomposition. That is, the original 3-D HSI is transformed into a 2-D matrix. We define the transformed 2-D HSI as $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_l] \in \mathbf{R}^{D\times L}$, where D is the spectral band, and L represents the total number of pixels in HSI. Similarly, the decomposed components from the 2-D matrix can be represented as: background $\mathbf{G} = [\mathbf{g}_1, \mathbf{g}_2, \ldots, \mathbf{g}_l] \in \mathbf{R}^{D\times L}$ and target $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_l] \in \mathbf{R}^{D\times L}$. The background and known target dictionaries are expressed as $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m] \in \mathbf{R}^{D\times M}$ and $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n] \in \mathbf{R}^{D\times N}$, respectively, where N and M are the number of dictionary atoms. The principle of DDNRTD is detailed as follows.

A. Proposed SSBD-Based Background Dictionary Construction

The background dictionary construction method directly determines the representability of the background part, which also further affects the target detection performance. In hyperspectral anomaly detection, background dictionary construction methods mostly rely on unsupervised clustering methods. For example, in [37] and [38], K-means clustering is combined with a simple and effective anomaly detection method, called RX [39], to construct the background dictionary. Although this method has the advantages of simplicity, the K value is usually set empirically and K-means is sensitive to anomalies and noise, which limits its practical applicability. In [40], the density peak clustering (DPC) method is used to extract relatively pure background pixels and strongly anomalous pixels, to construct background and anomaly dictionaries. Similarly, DPC is also affected by the setting of clustering parameters and suffers from the "curse of dimensionality" problem in hyperspectral data. Moreover, in some recent studies, generative adversarial networks have been widely examined for hyperspectral anomaly detection [41], [42], which use density-based spatial clustering to search potential background pixels.

In this article, we propose an SSBD-based background dictionary construction method for hyperspectral target detection, which integrates SAD and sparse representations theories. SSBD can be considered as a two-step structure. First, the SAD between the known target spectrum in target dictionary $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3]$ and the tested data $\mathbf{Y}$ is calculated based on the following equation:

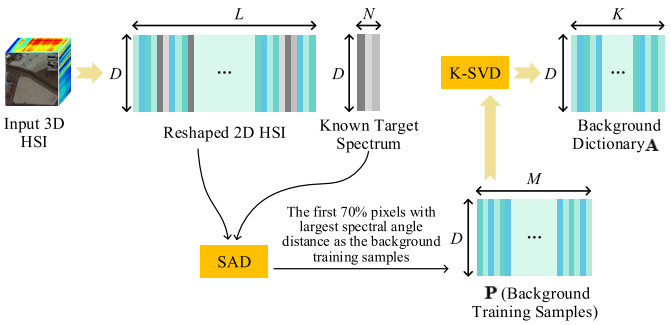


Fig. 2. Proposed SSBD-based background dictionary construction strategy.

$$D_{\mathrm{SAD}}(\mathbf{y}_i) = \cos^{-1}\frac{\mathbf{y}_i\mathbf{b}_j^{\mathrm{T}}}{\left(\mathbf{b}_j^{\mathrm{T}}\mathbf{b}_j\right)^{1/2}\left(\mathbf{y}_i^{\mathrm{T}}\mathbf{y}_i\right)^{1/2}} \tag{1}$$

where $\mathbf{y}_i$ represents the ith pixel vector in $\mathbf{Y}$, and $\mathbf{b}_j$ is the jth known target spectrum in $\mathbf{B}$. The spectral angle between the tested pixel and the known target spectrum is calculated one by one, and the average value is taken as the final result. The tested pixels with larger spectral angle are considered as background samples. Therefore, we choose the first 70% pixels with largest spectral angles as the training set $\mathbf{P}$. Following this, the K singular value decomposition (K-SVD) algorithm [43] is employed for $\mathbf{P}$ to train the background dictionary $\mathbf{A}$. The flowchart of the proposed SSBD-based dictionary construction strategy is shown in Fig. 2.

B. Proposed DDNRTD Method

The optimization objective function of the nonlinear representation model is expressed as

$$\begin{aligned}&\min_{\mathbf{X},\mathbf{Z}} \frac{1}{2}\|\mathbf{Y} - \mathbf{AX} - \mathbf{BZ}\|_F^2 + \|\mathbf{X}\|_{w*} + \lambda\|\mathbf{Z}\|_{2,1}\\ &\text{s.t. } \mathbf{Y} = \mathbf{AX} + \mathbf{BZ}\end{aligned} \tag{2}$$

where $\|\cdot\|_{w*}$ expresses the reweighted nuclear norm regularization, which is used to constrain background representation matrix $\mathbf{X}$. λ is a tradeoff parameter. Varying from the standard nuclear norm, the reweighted nuclear norm fully considers the contribution of different singular values to the matrix. Therefore, in the process of iteration, a weighting factor is added to balance the singular values. To be specific, $\mathbf{X}$ could be decomposed into three components, that is, $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$. Here, $\mathbf{U}$ and $\mathbf{V}$ are the left and right singular matrices, respectively. $\mathbf{\Sigma}$ is a diagonal matrix, and the ith diagonal element is Σ_{ii}. A weight factor $w_i = ((1/\Sigma_{ii}) + \varepsilon)$ is introduced to balance the singular values, where ε is a small positive parameter used to avoid negative values. The reweighted soft-threshold shrinkage is expressed as

$$G_w(\Sigma_{ii}) = \max(\Sigma_{ii} - \rho \times w_i, 0) \tag{3}$$

where ρ is a tradeoff parameter. Hence, in the iterative optimization, the larger singular value becomes even larger, while the smaller singular value approaches 0. The matrix $\mathbf{X}$ is then expressed as $\mathbf{X} = \mathbf{U}G_w(\Sigma_{ii})\mathbf{V}$.

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$\|\cdot\|_{2,1}$ is the $\ell_{2,1}-$ norm regularization, which is used to constrain the target representation matrix $\mathbf{Z}$. For $\mathbf{Z}$, the $\ell_{2,1}-$ norm is denoted as

$$\|\mathbf{Z}\|_{2,1}=\sum_{i=1}^{n}\sqrt{\sum_{j=1}^{t}\mathbf{z}_{i,j}^{2}}=\sum_{i=1}^{n}\left\|z_{i,:}\right\|_{2} \tag{4}$$

where i and j represent the numbers of row and column in $\mathbf{Z}$, respectively. The calculation of $\ell_{2,1}-$ norm could be regarded as first calculating the ℓ_2-norm of each row and then calculating the ℓ_1- norm of all columns. Auxiliary variables are introduced to make the objective optimization function in (2) separable

$$\begin{aligned}&\min_{\mathbf{X},\mathbf{Z},\mathbf{V}_1,\mathbf{V}_2}\|\mathbf{Y}-\mathbf{A}\mathbf{X}-\mathbf{B}\mathbf{Z}\|_F^2+\|\mathbf{V}_1\|_{w,*}+\lambda\|\mathbf{V}_2\|_{2,1}\\&\text{s.t. }\mathbf{V}_1=\mathbf{X},\quad\mathbf{V}_2=\mathbf{Z}.\end{aligned} \tag{5}$$

The augmented Lagrange function is expressed as follows:

$$\begin{aligned}&L\{\mathbf{X},\mathbf{Z},\mathbf{V}_1,\mathbf{V}_2,\mu\}\\&=\frac{1}{2}\|\mathbf{Y}-\mathbf{A}\mathbf{X}-\mathbf{B}\mathbf{Z}\|_F^2+\|\mathbf{V}_1\|_{w,*}\\&\quad+\lambda\|\mathbf{V}_2\|_{2,1}+\frac{\mu}{2}\|\mathbf{V}_1-\mathbf{X}-\mathbf{D}_1\|_F^2+\frac{\mu}{2}\|\mathbf{V}_2-\mathbf{Z}-\mathbf{D}_2\|_F^2\end{aligned} \tag{6}$$

where μ is a penalty parameter, and $\mathbf{D}_1$ and $\mathbf{D}_2$ are Lagrange multipliers. In this article, with an iterative process, the variables are optimized one by one, by fixing the other variables to minimize (6), thus obtaining the global optimum. Specifically, the optimization problem of objective function can be decomposed into the following subproblems.

1) Update $\mathbf{X}$: The optimization problem of $\mathbf{X}$ can be expressed as

$$\begin{aligned}\mathbf{X}^{t+1}=&\arg\min_{\mathbf{X}^t}\frac{1}{2}\left\|\mathbf{Y}-\mathbf{A}\mathbf{X}^t-\mathbf{B}\mathbf{Z}^t\right\|_F^2\\&+\frac{\mu}{2}\left\|\mathbf{V}_1^t-\mathbf{X}^t-\mathbf{D}_1^t\right\|_F^2\end{aligned} \tag{7}$$

where t is the iteration index. By deriving (4) for $\mathbf{X}^t$ and setting it to 0, $\mathbf{X}^{t+1}$ can be calculated as follows:

$$\mathbf{X}^{t+1}=(\mathbf{A}^{\mathrm{T}}\mathbf{A}+2\mu\mathbf{I})^{-1}\left[\mathbf{A}^{\mathrm{T}}(\mathbf{Y}-\mathbf{B}\mathbf{Z}^t)+\mu\left(\mathbf{V}_1^t-\mathbf{D}_1^t\right)\right]. \tag{8}$$

2) Update $\mathbf{V}_1$: The objective function of $\mathbf{V}_1$ is written as

$$\mathbf{V}_1^{t+1}=\arg\min_{\mathbf{V}_1^t}\left\|\mathbf{V}_1^t\right\|_{w,*}+\frac{\mu}{2}\left\|\mathbf{V}_1^t-\mathbf{X}^{t+1}-\mathbf{D}_1^t\right\|_F^2. \tag{9}$$

SVD is used to solve (9), and $\mathbf{V}_1^{t+1}$ can be calculated as follows:

$$\mathbf{V}_1^{t+1}=\Theta_{w,1/\mu^t}\left(\mathbf{X}^{t+1}+\mathbf{D}_1^t\right). \tag{10}$$

3) Update $\mathbf{V}_2$: The optimization problem of $\mathbf{V}_2$ is expressed as follows:

$$\mathbf{V}_2^{t+1}=\arg\min_{\mathbf{V}_2^t}\frac{\mu}{2}\left\|\mathbf{V}_2^t-\mathbf{X}^{t+1}-\mathbf{D}_2^t\right\|_F^2+\lambda\left\|\mathbf{V}_2^t\right\|_{2,1}. \tag{11}$$

Equation (8) can be solved using the $\ell_{2,1}-norm$ minimization operator Ω [44]. $\mathbf{V}_2^{t+1}$ can be calculated as follows:

$$\mathbf{V}_2^{t+1}=\Omega_{\lambda/\mu^t}\left(\mathbf{Z}^t+\mathbf{D}_2^t\right). \tag{12}$$

4) Update $\mathbf{Z}$: The optimization problem of $\mathbf{Z}$ is as follows:

$$\begin{aligned}\mathbf{Z}^{t+1}=&\arg\min_{\mathbf{Z}^t}\frac{1}{2}\left\|\mathbf{Y}-\mathbf{A}\mathbf{X}^t-\mathbf{B}\mathbf{Z}^t\right\|_F^2\\&+\frac{\mu}{2}\left\|\mathbf{V}_2^t-\mathbf{Z}^t-\mathbf{D}_2^t\right\|_F^2.\end{aligned} \tag{13}$$

Similarly, the update process of $\mathbf{X}$, $\mathbf{Z}^{t+1}$ can be calculated as follows:

$$\begin{aligned}&\mathbf{Z}^{t+1}\\&=(\mathbf{B}^{\mathrm{T}}\mathbf{B}+2\mu\mathbf{I})^{-1}\left[\mathbf{B}^{\mathrm{T}}\left(\mathbf{Y}-\mathbf{A}\mathbf{X}^{t+1}\right)+\mu\left(\mathbf{V}_2^t-\mathbf{D}_2^t\right)\right].\end{aligned} \tag{14}$$

5) Update the Lagrange multipliers $\mathbf{D}_1$ and $\mathbf{D}_2$

$$\begin{cases}\mathbf{D}_1^{t+1}=\mathbf{D}_1^t-\left(\mathbf{V}_1^{t+1}-\mathbf{X}^{t+1}\right)\\\mathbf{D}_2^{t+1}=\mathbf{D}_2^t-\left(\mathbf{V}_2^{t+1}-\mathbf{Z}^{t+1}\right).\end{cases} \tag{15}$$

6) Update the penalty parameter μ

$$\mu^{\mathrm{t}+1}=\min(\rho\mu^t,\mu_{\max}) \tag{16}$$

where ρ is the growth index of penalty parameter [as define in (3)], and $\mu_{\max}$ is the maximum penalty parameter.

The iteration process stops upon reaching the maximum iteration index or upon meeting the following requirements, where ε is the residual threshold:

$$\begin{cases}\|\mathbf{Y}-\mathbf{A}\mathbf{X}-\mathbf{B}\mathbf{Z}\|_F\le\varepsilon\\\|\mathbf{X}-\mathbf{V}_1\|_F\le\varepsilon\\\|\mathbf{Z}-\mathbf{V}_2\|_F\le\varepsilon.\end{cases} \tag{17}$$

Finally, the target part is represented by the product of the target dictionary and the corresponding representation coefficient, and the final detection map is produced by computing the ℓ_2- norm of each spectral vector of the target part. The process of solving the objective function in (2) is summarized in Algorithm 1. The complete flowchart of the DDNRTD algorithm is shown in Fig. 3.

III. Experiments

A. .Experimental Datasets and Setup

For validation of the proposed DDNRTD method, two simulated and five real-world hyperspectral datasets were used in the experiments. The background parts of the two simulated hyperspectral datasets are based on a real scene, which was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) from San Diego airport. The two background parts consist of 189 and 126 spectral bands after removal of the bands with a low signal-to-noise ratio (SNR), and the bands affected by water vapor absorption. Four target spectrums were selected from each of the San Diego beach and Pavia datasets. The San Diego beach and Pavia scenes were collected using the AVIRIS and Reflective Optics Spectrographic Imaging

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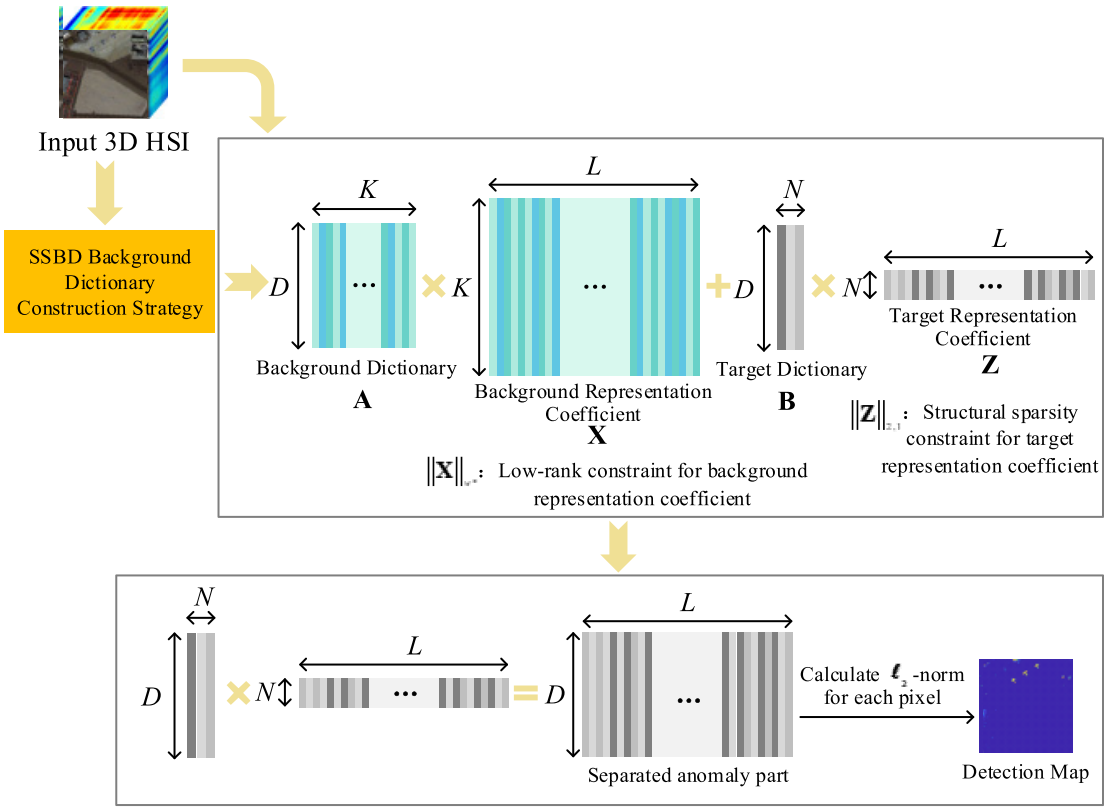


Fig. 3. Flowchart of the proposed DDNRTD method for hyperspectral subpixel target detection.

Algorithm 1 Optimization Procedure of the Proposed DDNRTD Method

1. **Input**: Two-dimensional HSI $\mathbf{Y}$, background dictionary $\mathbf{A}$, target dictionary $\mathbf{B}$, parameters λ, μ, $\mu_{\max}$ and ρ, and the maximum iteration index $t_{\max}$, and residual threshold ε.
2. **Output**: Background representation coefficient $\mathbf{X}^*$ and target representation coefficient $\mathbf{Z}^*$.
3. **Initialization**: Set $\mathbf{X}$, $\mathbf{Z}$, $\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{D}_1$, $\mathbf{D}_2$ to zero matrices.
4. **while** converge = false, **do**
5. Update $\mathbf{X}$ by Eq. (7).
6. Update $\mathbf{V}_1$ by Eq. (9).
7. Update $\mathbf{V}_2$ by Eq. (11).
8. Update $\mathbf{Z}$ by Eq. (13).
9. Update $\mathbf{D}_1$ and $\mathbf{D}_2$ by Eq. (15).
10. Update μ by Eq. (16).
11. **if** $\begin{cases} \|\mathbf{Y}-\mathbf{AX}-\mathbf{BZ}\|_F \le \varepsilon \\ \|\mathbf{X}-\mathbf{V}_1\|_F \le \varepsilon \\ \|\mathbf{Z}-\mathbf{V}_2\|_F \le \varepsilon \end{cases}$ or $t = t_{\max}$, converge = true.
12. **end if**
13. **end while**

System (ROSIS-03), containing 103 and 193 spectral bands, respectively. Subpixel targets were generated according to the target implantation method described in [45]. Specifically, a subpixel target $\mathbf{s}$ was simulated by mixing the background spectrum $\mathbf{g}$ and the target spectrum $\mathbf{t}$ with a certain proportion. Fig. 4 shows the background scene and target spectrum, as well as the false color images of the two simulated datasets and the corresponding ground-truth map. The first simulated dataset (i.e., simulated dataset I) consists of 103 spectral bands with a spatial size of 100 × 100 pixels. Note that the first 103 bands in the original San Diego airport dataset were selected to match that of the target spectra. Similarly, the second simulated dataset (i.e., simulated dataset II) consists of 126 spectral bands, with 100 × 100 pixels in each band. There are 36 target subpixels in each simulated dataset. Specifically, in each row, the target proportion is the same, which is 0.1, 0.2, 0.4 and 0.8 for rows from top to bottom. Correspondingly, the target spectrum are $\mathbf{t}_4$, $\mathbf{t}_1$, $\mathbf{t}_2$ and $\mathbf{t}_3$ (see the legend in Fig. 4) from top to bottom. Only $\mathbf{t}_2$ and $\mathbf{t}_4$ were chosen as the known target spectrum in the experiments.

Five real-world hyperspectral datasets were collected using different sensors. Table I lists some important image characteristics of the five real datasets. The false-color images (bands 37, 18, and 8 as RGB) and corresponding ground-truth maps are shown in Fig. 5. The subpixel targets are mainly located at the boundary of the entire object.

Six popular and state-of-the-art hyperspectral target detection methods were used as benchmarks, namely CEM [7], OSP [9], SRBBH [12], SLRMD [13], ACE [32], and AMSD [33]. Fig. 6 shows the curves of the known target spectra for the five real datasets. $\mathbf{t}_1$, $\mathbf{t}_2$, and $\mathbf{t}_3$ were collected from the real images and constitute the target dictionary, which were used as the input for DDNRTD, STD, SRBBH, and

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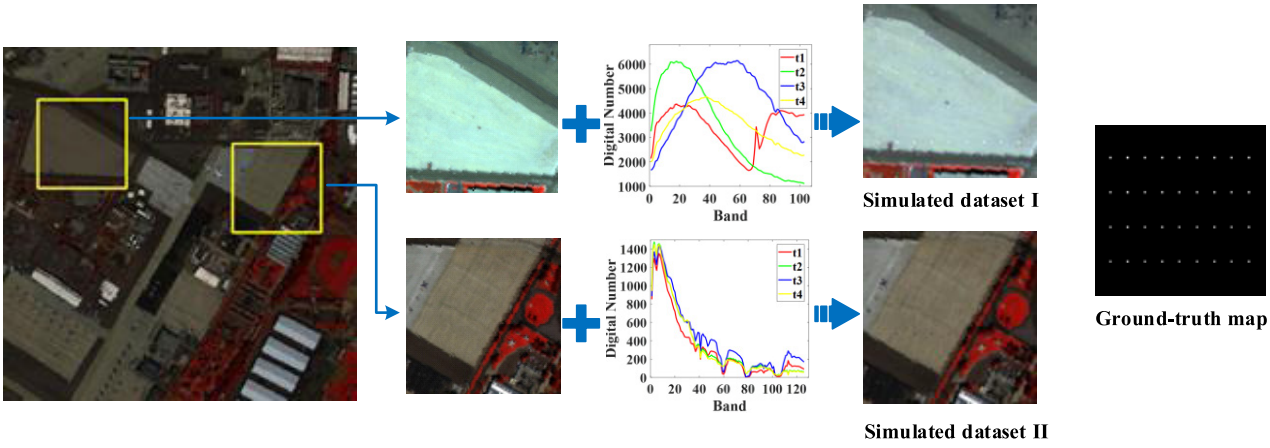


Fig. 4. Two simulated hyperspectral datasets.

TABLE I
SOME IMAGE CHARACTERISTICS OF THE FIVE REAL DATASETS

Datasets	Location	Sensor	Spatial size	No. of Bands
Dataset I	Sandiego	AVIRIS	150×150	189
Dataset II	Texas coast	AVIRIS	100×100	204
Dataset III	Sandiego	AVIRIS	100×100	193
Dataset IV	Hydice	HYDICE	80×120	175
Dataset V	Pavia	ROSIS-03	185×100	103

SLRMD. Additionally, we took the mean of $\mathbf{t}_1$, $\mathbf{t}_2$, and $\mathbf{t}_3$ as the input of single target spectrum for the remaining benchmark methods. The 3-D and 2-D receiver operating characteristic (ROC) curves [46], [47] and the area under the curve (AUC) values [48] were used as the evaluation indexes for experimental results. ROC curves show the tradeoff between detection probability and false alarm rate. Furthermore, there exist two versions of the 2-D ROC curves. In one version, the probability of false alarm (P_f) and the probability of detection (P_d) were used as abscissa and ordinate, respectively, while in the other version, thresholds τ and P_d were used as abscissa and ordinate, respectively. The AUC value is the area under the 2-D ROC curve, and the ideal values of AUC (P_f, P_d) and AUC (τ, P_f) are 1 and 0, respectively. In addition, Box–Whisker plot was also utilized to analyze the separability between background and targets.

The rest of this section consists of three parts. Sections III-B and III-C present the target detection results based on seven methods for the simulated and real datasets, respectively. Section III-D analyzes the influence of the tradeoff parameter on the performance of the proposed DDNRTD method.

B. Results for the Simulated Datasets

In the experiments, the dual-window sizes and sparsity level of SRBBH were set to (7, 5, 14) for the two simulated datasets. In SLRMD, the tradeoff parameters were set to $\lambda = 1$ and $\beta = 0.01$ for both datasets. The tradeoff parameters in DDNRTD were set to $\lambda = 0.005$, and the background dictionary size was 256 for both datasets. Based on the seven target detection methods, the target detection maps for the two simulated datasets are presented in Fig. 7. It is obvious that compared with the benchmark methods, DDNRTD can detect the subpixel targets more effectively, especially for subpixels with small target mixing proportions. For example, for the subpixel targets in the first row of simulated dataset I, CEM, OSP, AMSD, ACE, and SRBBH failed to detect almost all of them. Although SLRMD could detect all target subpixels, as shown in Fig. 7(g1) and (g2), DDNRTD could suppress background and reduce the false alarm rate more satisfactorily. The main reason for this is that DDNRTD imposes low-rank constraint on the representation matrix of background dictionary, rather than the original background part, like in case of SLRMD. As a result, the nonlinear representation model based on double dictionaries can restore the background part and extract the target part more effectively.

Fig. 8 shows the 3-D and 2-D ROC curves for the two simulated datasets based on the seven methods. It can be seen that the 2-D ROC curves (P_f, P_d) of DDNRTD are located in the top left corner in Fig. 8(a1) and (a2). In Fig. 8(b1) and (b2), the 2-D ROC curves (τ, P_f) also suggest that DDNRTD is a more reliable choice than most of the benchmark methods. Through the 3-D ROC curves, we can observe more intuitively that the proposed DDNRTD method is the most satisfactory method overall. Additionally, in Fig. 9, Box–Whisker plots show that DDNRTD is highly capable of separating the background and the target. More precisely, the detection results of DDNRTD present distance and smaller overlap between the background box and the target box.

The AUC values based on the seven target detection methods are listed in Table II, where the most accurate results of AUC(P_f, P_d) and AUC(τ, P_f) for each dataset are highlighted in boldface. It is seen clearly the advantages of DDNRTD over the benchmark methods. Taking the AUC(τ, P_f) values for simulated dataset I as an example, the AUC(τ, P_f) value of DDNRTD is 0.9991, which are 0.2103, 0.0460, 0.0860, 0.1295, 0.0281, and 0.0286 larger than those of CEM, OSP, AMSD, ACE, SRBBH, and SLRMD, respectively.

To test the robustness of DDNRTD against noise, the noises with different SNR values were added into the simulated dataset I. Specifically, the white Gaussian noises (with SNRs

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TABLE II
AUC VALUES $(P_d, P_f)/(P_f, \tau)$ OF THE SEVEN METHODS FOR THE TWO SIMULATED DATASETS

	CEM	OSP	AMSD	ACE	SRBBH	SLRMD	DDNRTD
Simulated dataset I	0.7888/0.3207	0.9531/0.2919	0.9131/0.0653	0.8696/**0.0160**	0.9710/0.0148	0.9705/0.1594	**0.9991**/0.0363
Simulated dataset II	0.9804/0.2351	0.9566/0.2704	0.9495/**0.0078**	0.9154/0.0089	0.9432/0.0816	0.8603/0.1788	**0.9949**/0.0144

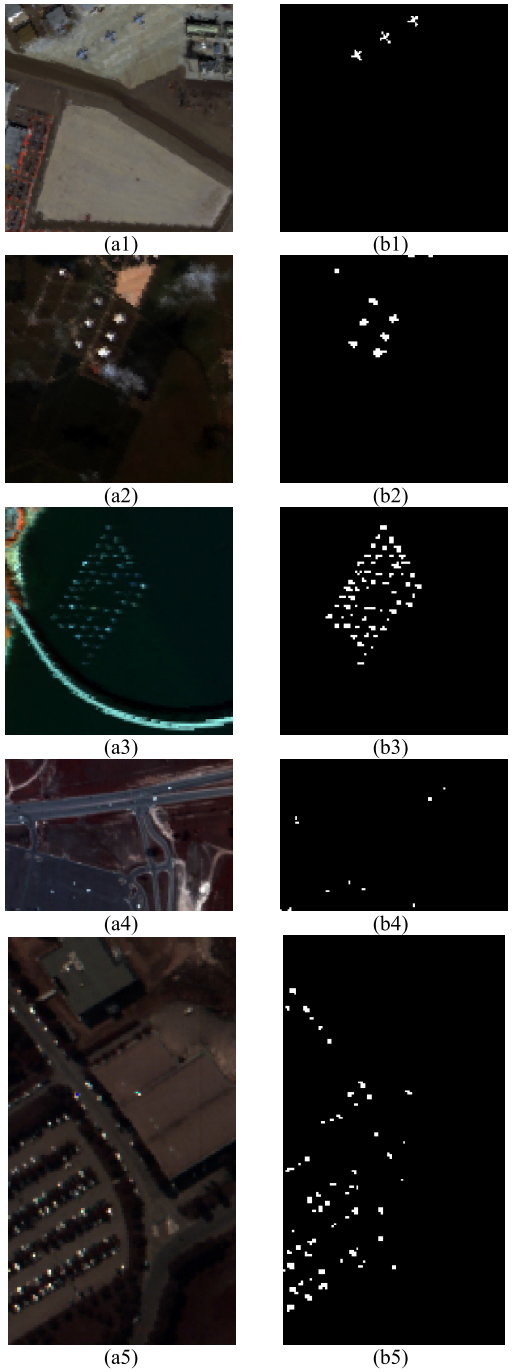


Fig. 5. Five real HSI datasets used in the experiments: (a1)–(a5) false-color images (bands 37, 18, and 8 as RGB) of the five real HSI datasets and (b1)–(b5) corresponding ground-reference maps.

of 35, 40, and 45 dB) were considered. Fig. 10 shows the false-color images of the simulated dataset I with different levels of white Gaussian noises, and Fig. 11 shows the corresponding target detection maps. It can be seen from Fig. 10 that when the SNR is 35 dB, there are noticeable noises in the target detection maps by all methods. However, compared with other benchmark methods, DDNRTD is still more robust with respect to the noises. Table III lists the corresponding AUC values of the seven target detection methods. The results are consistent with those depicted in Fig. 11.

C. Results for the Real Datasets

For the five real datasets (i.e., datasets I–V), the dual-window sizes and sparsity levels of SRBBH were set to (7, 5, 14), (7, 5, 12), (7, 5, 10), (9, 7, 12), and (7, 5, 20), respectively. The tradeoff parameters in SLRMD were set to $\lambda = 1$ and $\beta = 0.01$ for all five datasets. The tradeoff parameterλ in DDNRTD was set to 0.5, 0.005, 0,002, 0.002, and 0.002 for datasets I–V, respectively. The background dictionary sizes for datasets I and V were 256 and 1024, respectively, while those for the others was 512.

Fig. 12 shows the detection maps based on the seven target detection methods for the five real datasets. It can be seen that, DDNRTD not only detects almost all target pixels, but also effectively suppresses background interference. However, the benchmark methods cannot accomplish this. For example, in dataset III, where the size of targets is smaller, there are more subpixels at the boundary of each target. As seen in Fig. 12(a3)–(g3), CEM, AMSD, ACE, and SRBBH cannot detect most of the target pixels, especially the subpixels at the boundary, which makes the shape of the targets incomplete. SLRMD can accurately detect the position of the target, but the boundary is still inaccurate. As another example, in dataset V, which has complex background distribution, it is difficult to suppress the background and highlight the targets based on the benchmark methods. As seen in Fig. 12(a5)–(g5), CEM, AMSD, ACE, and SRBBH fail to detect the target pixels. Although SLRMD shows relatively stable performance, when compared with DDNRTD, the ability to suppress background is inferior.

The 3-D and 2-D ROC curves for the five datasets are shown in Fig. 13, and the Box–Whisker plots are shown in Fig. 14. Through the ROC curves and Box–Whisker plots, we can see that DDNRTD has excellent performance in target detection and suppression of false alarm. In addition, the corresponding AUC values are listed in Table IV, where the most accurate results of AUC(P_f, P_d) and AUC(τ, P_f) are highlighted in boldface. In agreement with the above analysis, DDNRTD produces the most accurate results (i.e., the largest AUC values) for all five datasets.

D. Parameter Analysis

The background dictionary size K is an important parameter in the proposed SSBD-based background dictionary

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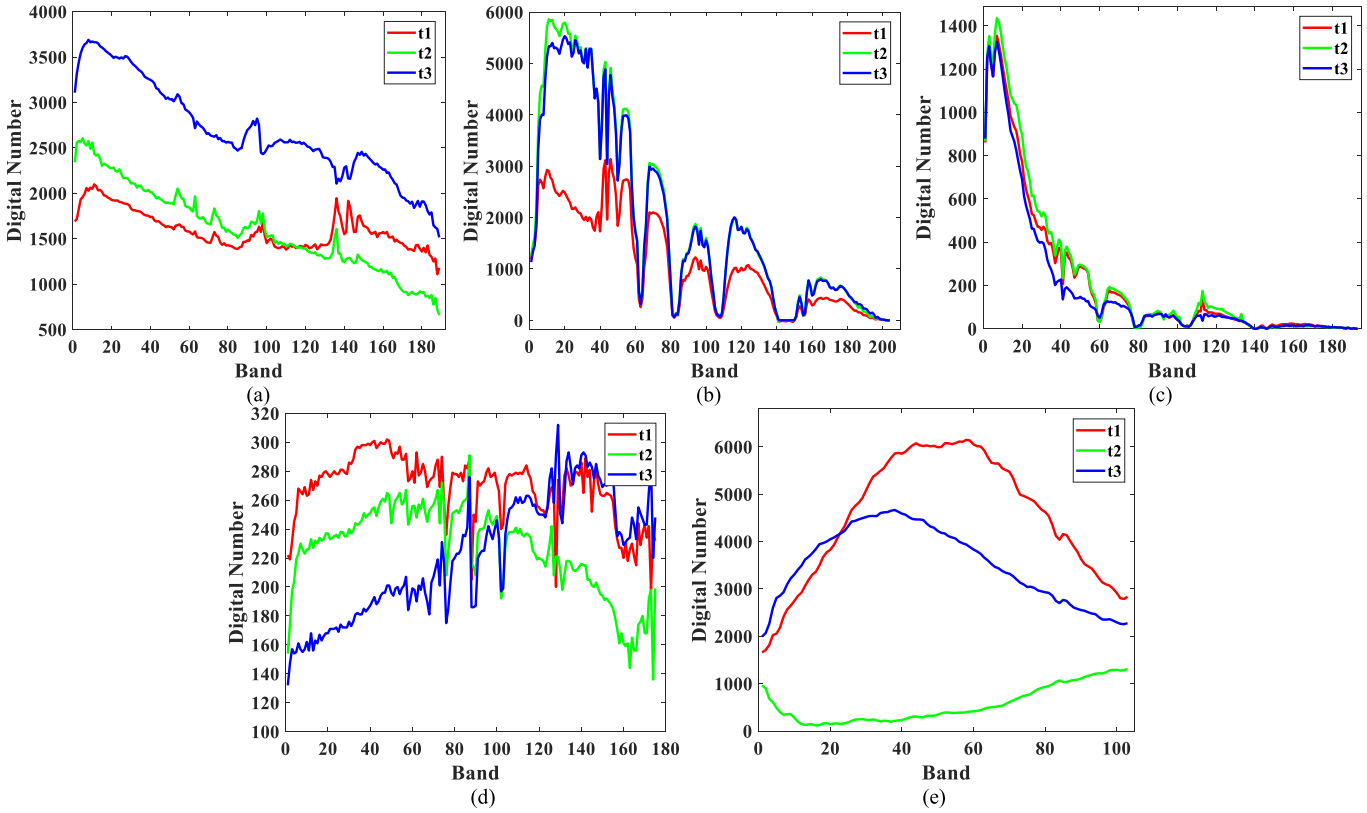


Fig. 6. Curves of known target spectra for the five real datasets: (a) dataset I; (b) dataset II; (c) dataset III; (d) dataset IV; and (e) dataset V.

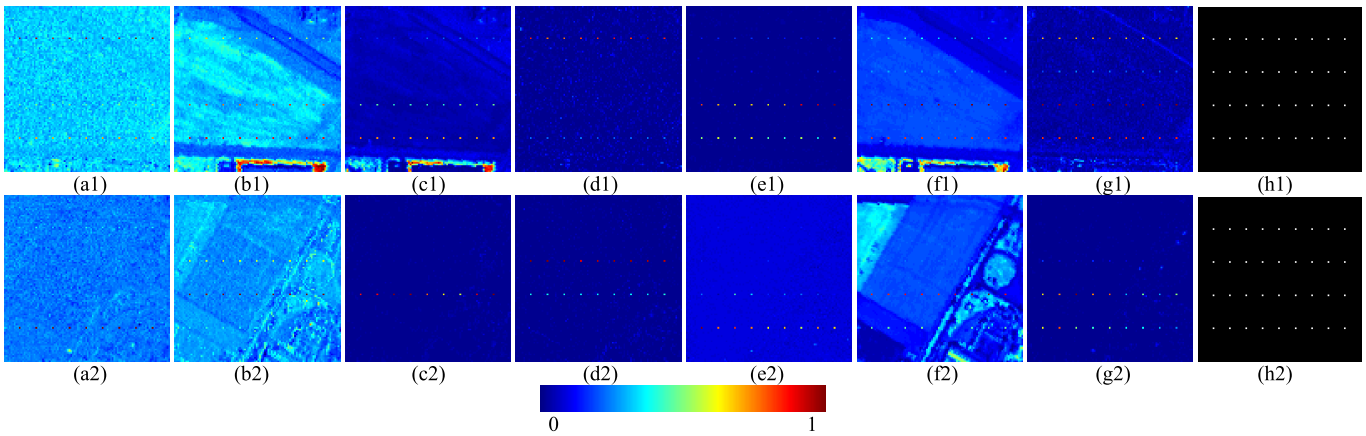


Fig. 7. Subpixel target detection maps based on seven methods for the two simulated datasets: (a)–(g) detection maps for CEM, OSP, AMSD, ACE, SRBBH, SLRMD, and DDNRTD and (h) corresponding ground-reference map. Lines 1 and 2 are results for simulated datasets I and II, respectively.

TABLE III

AUC VALUES $(P_d, P_f)/(P_f, \tau)$ OF THE SEVEN METHODS FOR THE SIMULATED DATASET I WITH DIFFERENT LEVELS OF WHITE GAUSSIAN NOISES

	CEM	OSP	AMSD	ACE	SRBBH	SLRMD	DDNRTD
35 dB	0.9600/0.3213	0.8038/0.3793	0.8422/**0.0160**	0.9749/0.0206	0.9531/0.0283	0.9671/0.1583	**0.9819**/0.0655
40 dB	0.9786/0.3599	0.7832/0.4540	0.8476/**0.0069**	0.9562/0.0161	0.9687/0.0447	0.9699/0.1604	**0.9957**/0.0486
45 dB	0.8998/0.2646	0.8093/0.4139	0.8693/**0.0078**	0.9159/0.0123	0.9770/0.0242	0.9705/0.1589	**0.9986**/0.0409

construction strategy. In general, K was set to an integer power of 2, and an important condition for constructing a complete dictionary is that the dictionary size needs to be larger than the dimension of the dictionary atom [47]. To analyze the influence of K value on the results, for all datasets used in the experiments, the K values of 256,

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TABLE IV
AUC VALUES $(P_d, P_f)/(P_f, \tau)$ OF THE SEVEN METHODS FOR THE FIVE REAL DATASETS

	CEM	OSP	AMSD	ACE	SRBBH	SLRMD	DDNRTD
Dataset I	0.9925/0.2756	0.9937/0.4166	0.9752/0.0107	0.9957/**0.0086**	0.7354/0.0671	0.9944/0.1358	**0.9992**/0.0563
Dataset II	0.8427/0.1569	0.9409/0.2277	0.9365/**0.0061**	0.8835/0.0067	0.9785/0.8445	0.9917/0.0899	**0.9985**/0.0091
Dataset III	0.8074/0.3990	0.8892/0.4296	0.6795/0.0371	0.7222/**0.0129**	0.9112/0.6761	0.8818/0.0665	**0.9855**/0.0176
Dataset IV	0.8662/0.2056	0.8695/0.2892	0.8425/0.0116	0.9470/**0.0104**	0.8749/0.8651	0.9051/0.0850	**0.9954**/0.0153
Dataset V	0.6994/0.3240	0.7301/0.3037	0.7292/0.0161	0.7601/0.0147	0.9311/0.0825	0.9863/0.0704	**0.9919/0.0073**

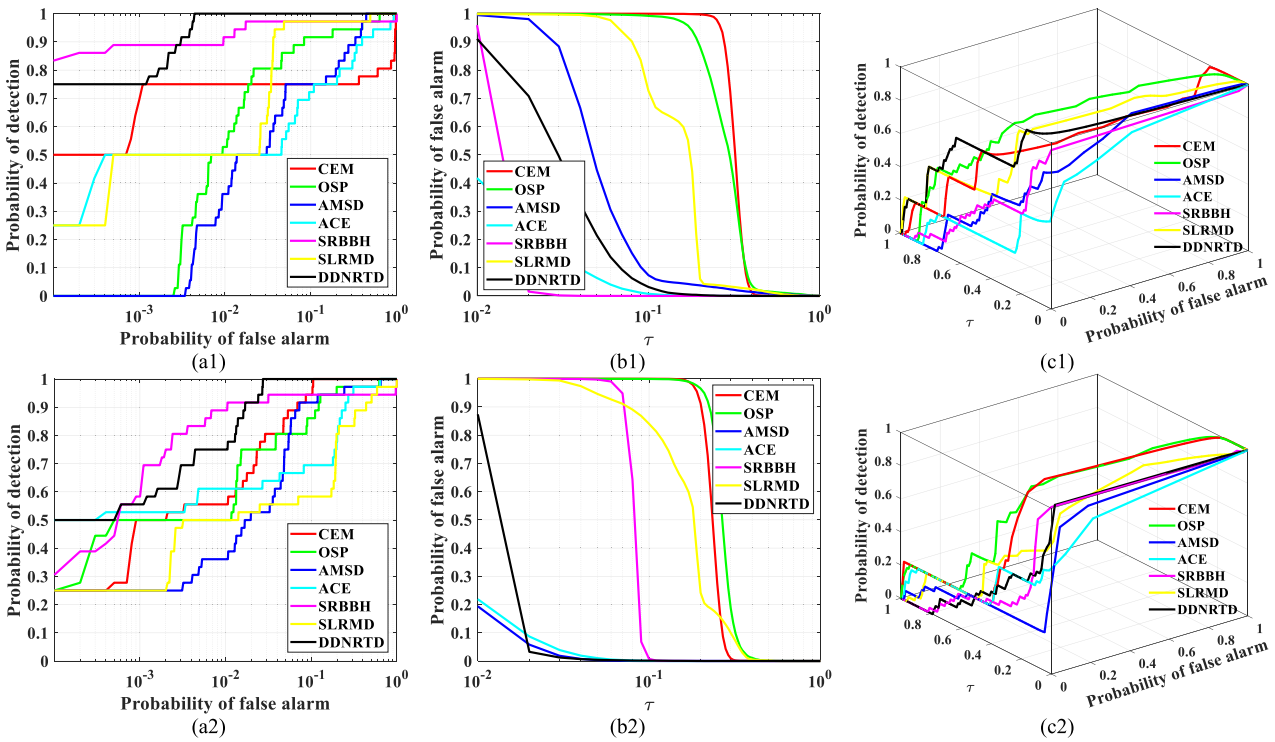


Fig. 8. ROC curves of the seven methods for the two simulated datasets. From left to right are: (a) 2-D ROC curves of (P_d, P_f); (b) 2-D ROC curves of (P_f, τ); and (c) 3-D ROC curves. Lines 1 and 2 are results for simulated datasets I and II, respectively.

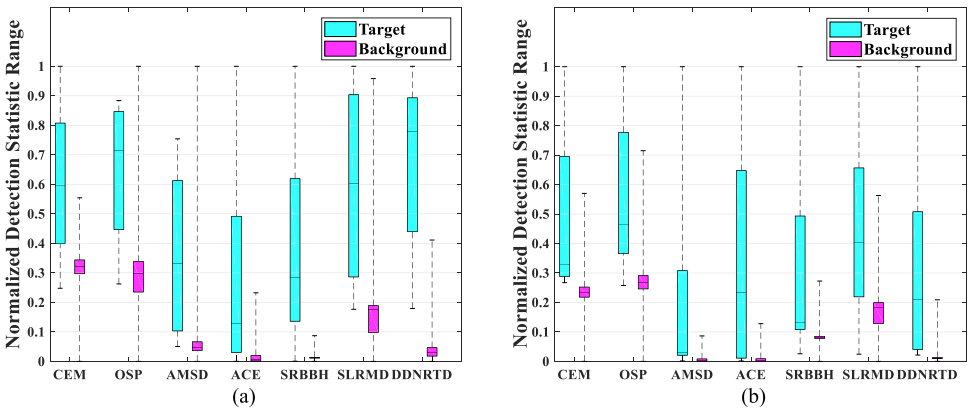


Fig. 9. Box–Whisker plots for the seven target detection methods for the two simulated datasets: (a) simulated dataset I and (b) simulated dataset II.

512, and 1024 were examined. For the proposed DDNRTD method, there is only one tradeoff parameter λ for the implicit regulation in target representation matrix $\mathbf{Z}$. To analyze the sensitivity of DDNRTD to the tradeoff parameter, the AUC values for different datasets were calculated by adjusting λ. Figs. 15 and 16 show the change in the AUC values for the

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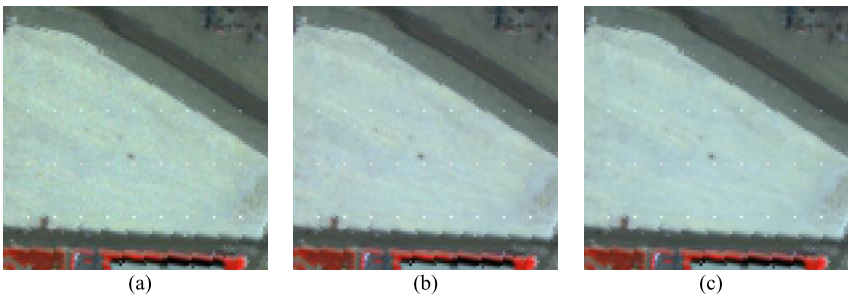


Fig. 10. False-color images (bands 37, 18, and 8 as RGB) of the simulated dataset I with different levels of white Gaussian noises: (a) 35 dB; (b) 40 dB; and (c) 45 dB.

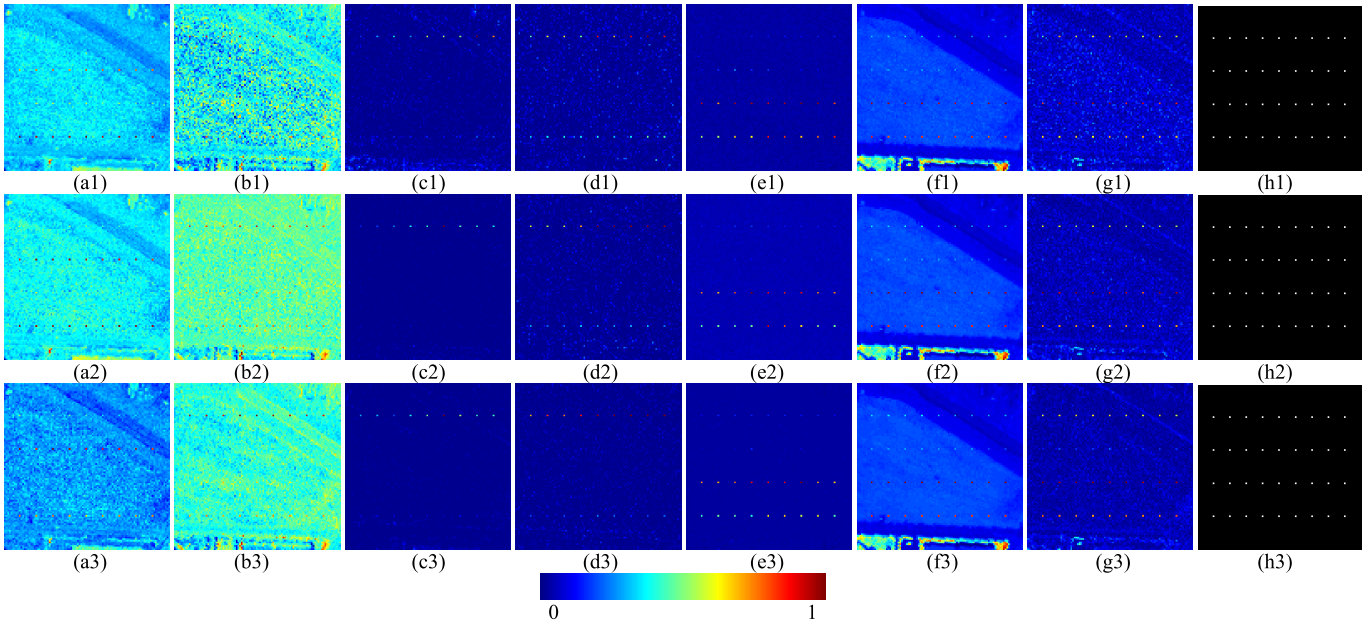


Fig. 11. Subpixel target detection maps based on seven methods for the simulated dataset I with different levels of white Gaussian noises: (a)–(g) detection maps for CEM, OSP, AMSD, ACE, SRBBH, SLRMD, and DDNRTD and (h) corresponding ground-reference map. Lines 1–3 are results for 35, 40, and 45 dB, respectively.

two simulated and five real datasets with the change in K andλ. Fig. 15 shows that in the simulated datasets, when $\lambda < 0.05$, there is no significant change in the AUC value under different K values. From Fig. 16, it is obvious that for real datasets I and V, when $\lambda = 0.02$ and $\lambda = 0.05$, respectively, the AUC values decrease significantly, but there are no significant changes under the other values of λ. Specifically, when the K value is 512, under various values of λ, the changes in the AUC values of the five real datasets are below 0.1285, 0.0387, 0.0571, 0.1525, and 0.1585, respectively. In short, the proposed DDNRTD method has a strong robustness to the change of tradeoff parameter λ for different datasets. Additionally, it can be seen that under the same λ, different background dictionary sizes can produce different AUC values, but the change trend of AUC values under different dictionary sizes are similar. When λ is smaller than 0.002, there is small difference between the AUC values produced by the three dictionary sizes.

IV. Discussion

Similar to the proposed DDNRTD, SLRMD is also derived from the LSRMD model. However, the experimental results show that DDNRTD is more adaptable to different datasets and consistently more accurate than SLRMD. For example, Tables II and III show that the AUC values of DDNRTD are much larger than those of SLRMD. In addition, compared with SLRMD, DDNRTD is very capable of suppressing background and reducing false alarm rate. For example, for the real dataset I in Fig. 9, the background contour can hardly be seen from the detection map based on DDNRTD, and there are obvious differences in brightness between background and targets. There are two main reasons for these results. First, SLRMD only uses a target dictionary to represent the target part and the sparsity constraint is applied to the target representation matrix, but the low-rank constraint is directly applied to the original background part. In DDNRTD, however, we use the double dictionaries scheme and impose constraints on their respective representation matrices. To some extent, the nonlinear representation model based on double dictionaries is more effective in decomposing hyperspectral data. Second, for the low-rank constraint of background part, SLRMD adopts the standard nuclear norm, while DDNRTD adopts the reweighted nuclear norm. Different from the standard nuclear norm, the reweighted nuclear norm considers the influence

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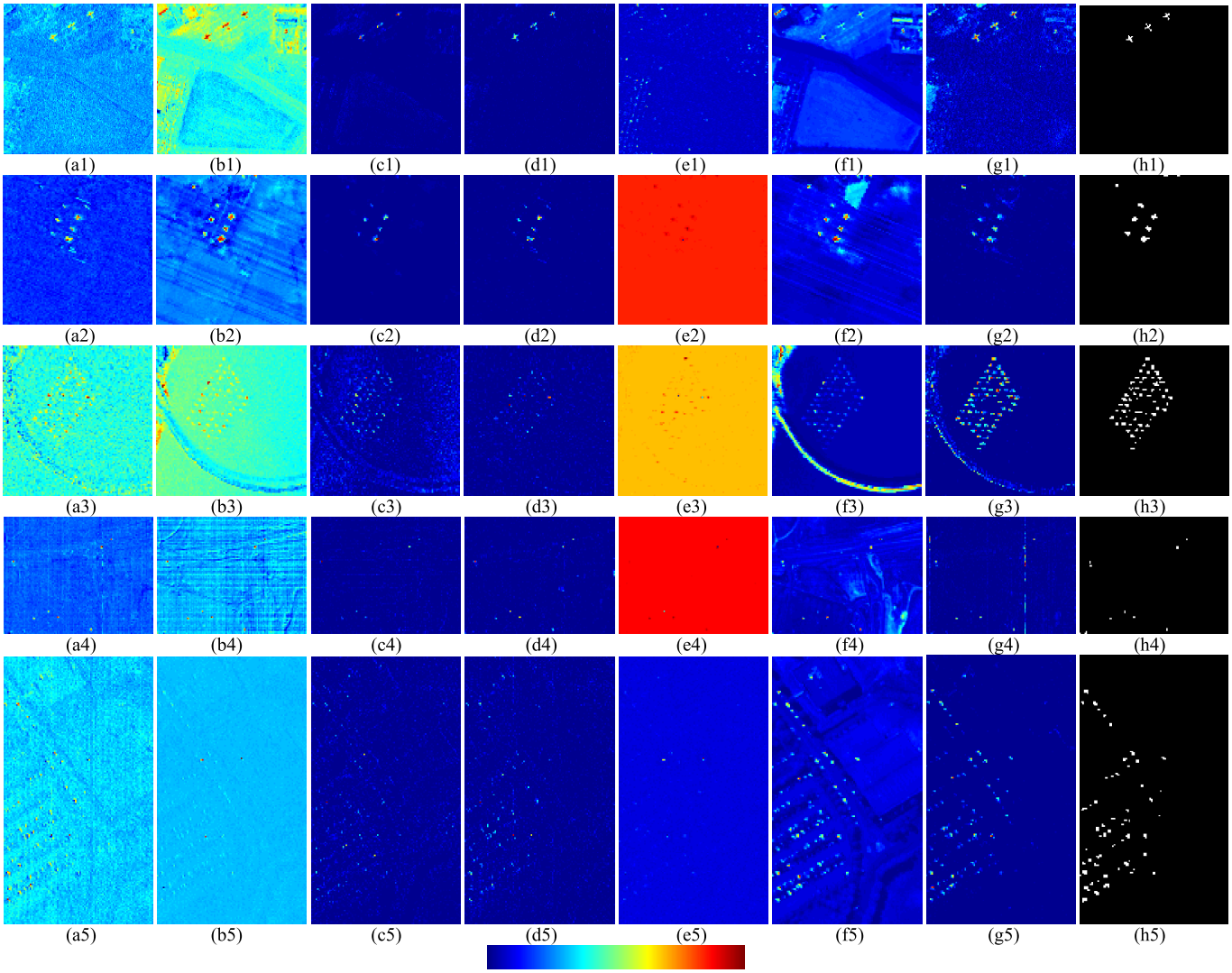


Fig. 12. Target detection maps based on seven methods for the five real datasets: (a)–(g) detection maps for CEM, OSP, AMSD, ACE, SRBBH, SLRMD, and DDNRTD and (h) corresponding ground-reference map. Lines 1–5 are results for datasets I–V, respectively.

of different singular values on the matrix, which brings the separated background closer to the real background.

Hyperspectral target detection still suffers from some limitations in practical applications. For example, it is difficult to obtain *a priori* spectrum. Moreover, due to the influence of atmospheric condition, the spectral reflectance of the target collected from the known HSI can be substantially different from that in the HSI for target detection, as the two HSIs can be acquired in very different conditions. Therefore, if there is no need for *a priori* spectrum in hyperspectral target detection, then its application value increases. In future research, it would be interesting to design an anomaly dictionary construction model and adapt DDNRTD to hyperspectral anomaly detection. This will not only overcome the limitations of the lack of *a priori* spectrum, but also improve the efficiency of subpixel anomaly detection.

Section III-B examines the robustness of different methods to the white Gaussian noise. From Fig. 11 and Table III, it is seen that DDNRTD is more robust. Under different levels of noise, DDNRTD can still produce more accurate detection results than other methods. However, there is an interesting phenomenon. Compared Fig. 7 with Fig. 11, and Table II with Table III, it is found that some benchmark methods could produce more accurate AUC values when the white Gaussian noises are involved. For example, based on ACE, the AUC values of simulated dataset I with 0, 35, 40, and 45 dB are 0.8696, 0.9749, 0.9562, and 0.9159, respectively. The main reason for this phenomenon is that the basic assumption of ACE is that the background distribution obeys normal distribution. When adding the white Gaussian noise into HSIs, the accuracy may not decrease but increase instead. This is not the case for methods without assumption on background distribution. For example, for SLRMD and DDNRTD, as the noise level drops from 45 to 35 dB, the AUC values decreases accordingly. In practical applications, most of the noises are more complex than the Gaussian distribution. Therefore, in future research, we should not only identify more appropriate models for charactering dataset distribution, but also for charactering noise distribution, to reduce the influence of noise in subpixel target detection.

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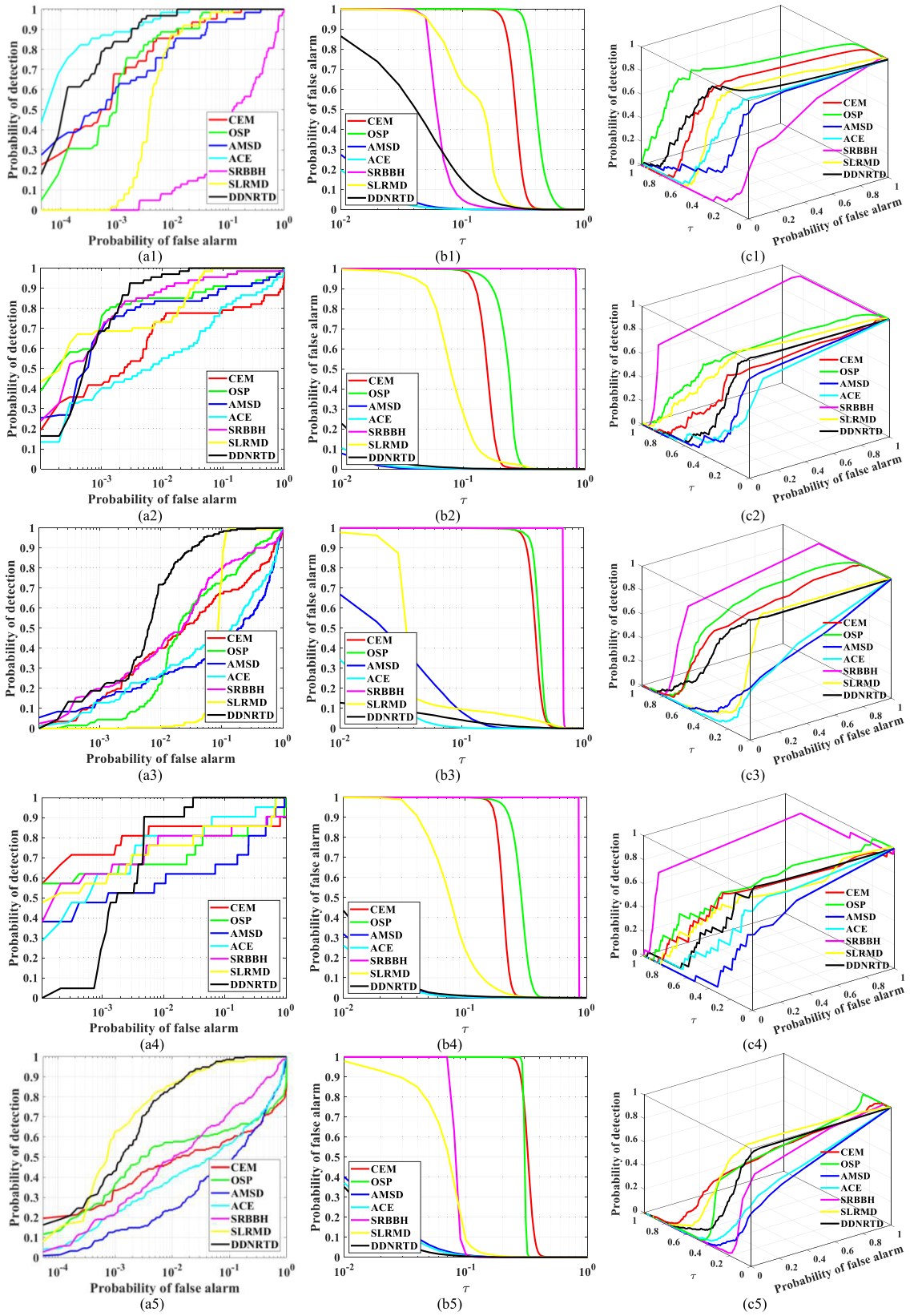


Fig. 13. ROC curves of the seven methods for the five real datasets. From left to right are: (a) 2-D ROC curves of (P_d, P_f); (b) 2-D ROC curves of (P_f, τ); and (c) 3-D ROC curves. Lines 1–5 are results for datasets I–V, respectively.

Section III-D demonstrates the robustness of DDNRTD with respect to the tradeoff parameter λ. From Figs. 15 and 16, we can see that when the range of λ is from 0.00001 to 0.002, for most datasets, the AUC values show no obvious change. When the range of λ increases further from 0.005 to 0.8, however, the AUC values of different datasets decrease by varying degrees. This indicates that a smaller λ can lead to satisfactory detection performance for different datasets.

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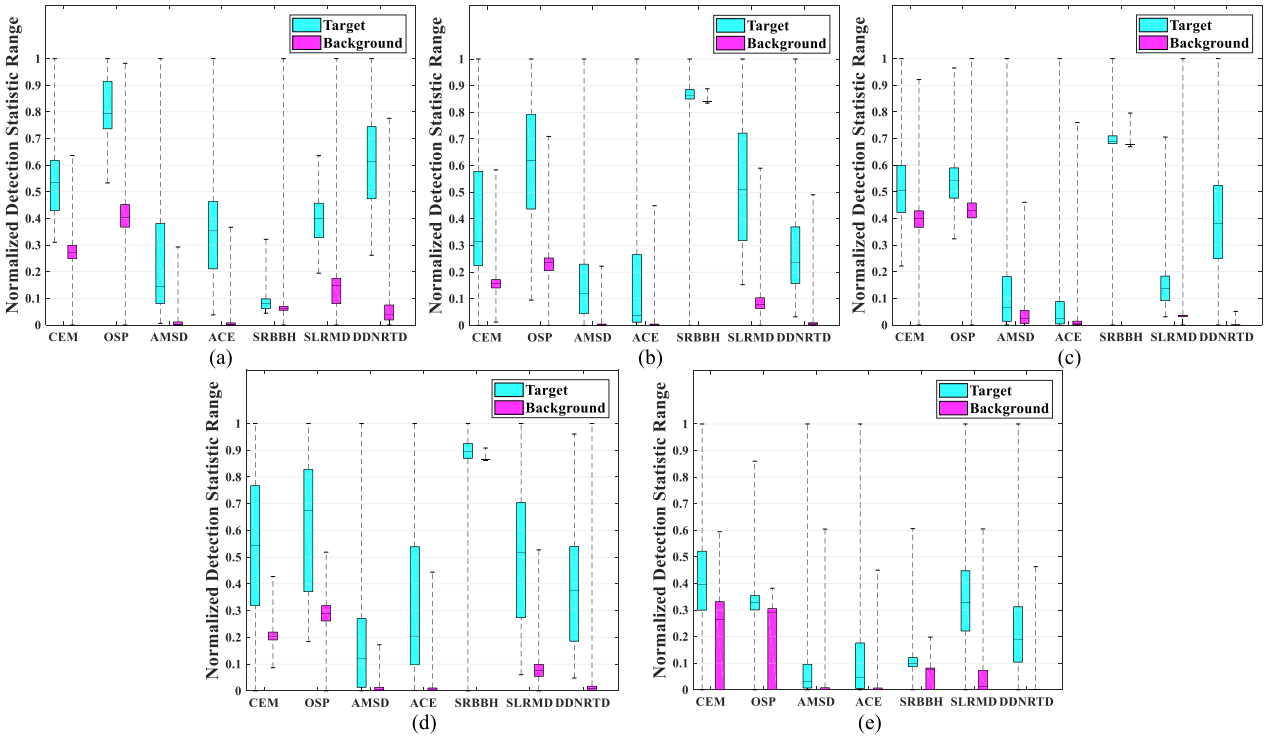


Fig. 14. Box–Whisker plots for the seven target detection methods for the five real datasets: (a) dataset I; (b) dataset II; (c) dataset III; (d) dataset IV; and (e) dataset V.

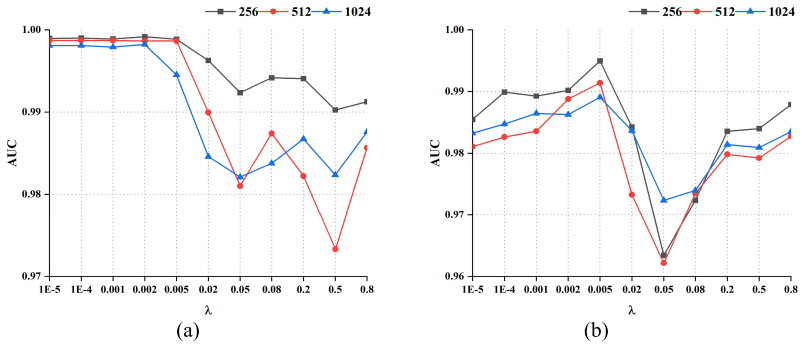


Fig. 15. AUC of the proposed DDNRTD under various background dictionary sizes and tradeoff parameters for the two simulated datasets: (a) simulated dataset I and (b) simulated dataset II.

In addition, in Fig. 16(d) and (e), it can be seen that when λ increases further, the AUC values decrease significantly. This is because there are many smaller targets in real datasets IV and V. Therefore, compared with large targets, small targets are more sensitive to larger λ values.

In this article, we construct simulated data based on the widely used linear mixing model. However, in practical cases, most of the mixed pixels in HSIs are mixed with complex nonlinearity. In addition, in the process of image acquisition, due to the point spread function (PSF) effect of sensors and many other factors (e.g., sunlight and atmosphere conditions) [50], the signal of pixel is always influenced by those of neighboring pixels. Therefore, in future research, to examine the robustness of DDNRTD under various imaging conditions, more factors will be considered in simulation of subpixel targets in HSIs. Correspondingly, it would also be worthwhile to develop hyperspectral subpixel target detection methods to fully consider the nonlinear properties of mixed pixels as well as the PSF effect.

Due to the effects of illumination and altitude of the platform, the acquired HSIs commonly contain the same objects with different spectrum and different objects with the same spectrum. This phenomenon of spectral variation deteriorates the detection performance obviously. In this article, the tested image was matched with *a priori* target spectrum in the background dictionary construction method. Due to the spectral variation, the matching results may not accord well with the real situation. The background dictionary may contain a reliable target spectrum, or may lose important background spectrum. The spectral variation is also an important issue in various remote-sensing image processing tasks, such as image classification [51], [52] and spectral unmixing. Overall,

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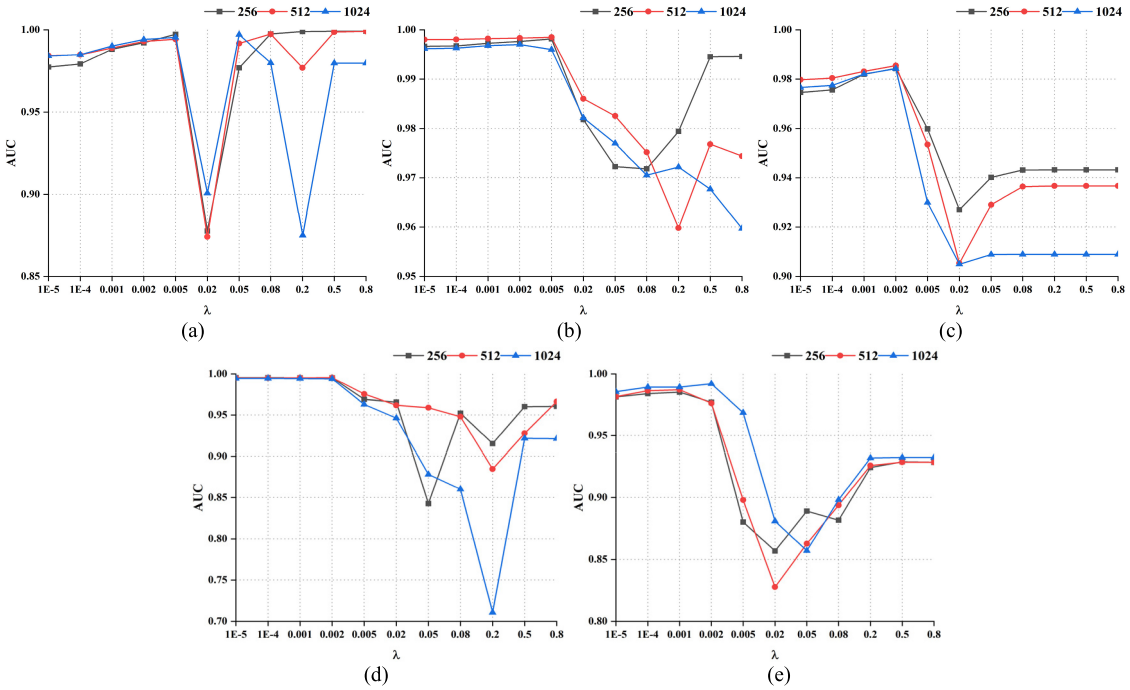


Fig. 16. AUC of the proposed DDNRTD under various background dictionary sizes and tradeoff parameters for the five real datasets: (a) dataset I; (b) dataset II; (c) dataset III; (d) dataset IV; and (e) dataset V.

substantial efforts need to be made to cope with this issue in hyperspectral subpixel target detection.

The curse of dimensionality is a common problem in HSI processing. Hence, as an important means to solve this problem, dimensionality reduction has been widely used as a preprocessing. In [53], to address the curse of dimensionality for hyperspectral classification, the graph embedding was used to extract features and map the original HSI into low-dimensional feature space. Considering the complementarity of different features, Luo *et al.* [54] proposed a multistructure unified discriminative embedding method to enhance the difference between different classes. Through extensive experiments, Kwan *et al.* [55] confirmed that fewer spectral bands could produce greater classification accuracy. Hence, in future research, it would be interesting to focus on the effect of band selection on detection performance. Additionally, in [55], some popular target and anomaly detection methods were extended for land cover classification based on HSIs. It would also be worthwhile to examine if DDNRTD can be extended to land cover classification.

V. Conclusion

In this article, we proposed the DDNRTD method for more accurate subpixel target detection. Moreover, in DDNRTD, we proposed the SSBD-based background dictionary construction strategy, which represents the background more effectively and increases the separation between background and target. Experiments on both simulated and real datasets were performed to validate the DDNRTD method. The results including target detection maps, Box–Whisker plots of the target and the background, and the ROC curves coupled with AUC showed that the proposed DDNRTD method is a satisfactory solution for hyperspectral subpixel target detection, which is more accurate than six benchmark methods.

References

[1] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5071–5084, Jun. 2021.

[2] F. Vincent and O. Besson, "One-step generalized likelihood ratio test for subpixel target detection in hyperspectral imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4479–4489, Jun. 2020.

[3] X. Shang *et al.*, "Target-constrained interference-minimized band selection for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6044–6064, Jul. 2021.

[4] X. Sun *et al.*, "Target detection through tree-structured encoding for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4233–4249, Sep. 2020.

[5] F. A. Kruse *et al.*, "The spectral image processing system (SIPS)-interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 145–163, May 1993.

[6] N. M. Nasrabadi, "Regularized spectral matched filter for target recognition in hyperspectral imagery," *IEEE Signal Process. Lett.*, vol. 15, pp. 317–320, 2008.

[7] W. H. Farrand and J. C. Harsanyi, "Mapping the distribution of mine tailings in the Coeur d'Alene river valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.

[8] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.

[9] C.-I. Chang, "Orthogonal subspace projection (OSP) revisited: A comprehensive study and analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 502–518, Mar. 2005.

[10] Q. Du, H. Ren, and C.-I. Chang, "A comparative study for orthogonal subspace projection and constrained energy minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1525–1529, Jun. 2003.

[11] C.-I. Chang and H. Ren, "An experiment-based quantitative and comparative analysis of target detection and image classification algorithms for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 2, pp. 1044–1063, Mar. 2000.

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[12] Y. Zhang, B. Du, and L. Zhang, "A sparse representation-based binary hypothesis model for target detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1346–1354, Mar. 2015.

[13] A. W. Bitar, L.-F. Cheong, and J.-P. Ovarlez, "Sparse and low-rank matrix decomposition for automatic target detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5239–5251, Aug. 2019.

[14] Y. Qu *et al.*, "Anomaly detection in hyperspectral images through spectral unmixing and low rank decomposition," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1855–1858.

[15] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[16] W. Xie *et al.*, "Unsupervised spectral mapping and feature selection for hyperspectral anomaly detection," *Neural Netw.*, vol. 132, pp. 144–154, Dec. 2020.

[17] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2020.

[18] A. Ma *et al.*, "Fast sequential feature extraction for recurrent neural network-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5920–5937, Jul. 2021.

[19] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, doi: 10.1109/TGRS.2021.3104032.

[20] Y. Shi, J. Li, Y. Li, and Q. Du, "Sensor-independent hyperspectral target detection with semisupervised domain adaptive few-shot learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6894–6906, Aug. 2021.

[21] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[22] X. R. Li, R. S. Huang, and L. Zhao, "Correntropy-based spatial–spectral robust sparsity-regularized hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jun. 2021.

[23] J. Qin *et al.*, "Blind hyperspectral unmixing based on graph total variation regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3338–3351, Apr. 2021.

[24] Q. Wang, C. Zhang, X. Tong, and P. M. Atkinson, "General solution to reduce the point spread function effect in subpixel mapping," *Remote Sens. Environ.*, vol. 251, Dec. 2020, Art. no. 112054.

[25] Q. Wang, C. Zhang, and P. M. Atkinson, "Sub-pixel mapping with point constraint," *Remote Sens. Environ.*, vol. 244, Jul. 2020, Art. no. 111817.

[26] P. Wang, L. Wang, H. Leung, and G. Zhang, "Super-resolution mapping based on spatial–spectral correlation for spectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2256–2268, Mar. 2021.

[27] X. Xu, J. Li, S. Li, and A. Plaza, "Sub-pixel component analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5564–5579, Apr. 2019.

[28] Y. Li, T. Lu, and S. Li, "Subpixel-pixel-superpixel-based multiview active learning for hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4976–4988, Jul. 2020.

[29] O. Duran and M. Petrou, "Spectral unmixing with negative and superunity abundances for subpixel anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 1, pp. 152–156, Jan. 2009.

[30] S. Khazai, A. Safari, B. Mojaradi, and S. Homayouni, "An approach for subpixel anomaly detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 769–778, Apr. 2013.

[31] Z. Huang, L. Fang, and S. Li, "Subpixel-pixel-superpixel guided fusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 5998–6007, Sep. 2020.

[32] S. Kraut, L. L. Scharf, and R. W. Butler, "The adaptive coherence estimator: A uniformly most-powerful-invariant adaptive detection statistic," *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 427–438, Feb. 2005.

[33] D. Manolakis, C. Siracusa, and G. Shaw, "Hyperspectral subpixel target detection using the linear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 7, pp. 1392–1409, Jul. 2001.

[34] J. J. Broadwater and R. Chellappa, "Hybrid detectors for subpixel targets," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 1, pp. 1891–1903, Nov. 2007.

[35] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.

[36] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, Jul. 2011.

[37] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[38] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep Plug-and-Play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021, doi: 10.1109/TGRS.2021.3049224.

[39] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[40] T. Cheng and B. Wang, "Total variation and sparsity regularized decomposition model with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1472–1486, Feb. 2021.

[41] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[42] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[43] M. Aharon, M. Elad, and A. Bruckstein, "*rmK*-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Trans. Signal Process.*, vol. 54, no. 11, pp. 4311–4322, Nov. 2006.

[44] G. Liu, Z. Lin, S. Yan, J. Sun, Y. Yu, and Y. Ma, "Robust recovery of subspace structures by low-rank representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 171–184, Jan. 2013.

[45] S. M. Schweizer and J. M. F. Moura, "Efficient detection in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 10, no. 4, pp. 584–597, Apr. 2001.

[46] M. Song, X. Shang, and C.-I. Chang, "3-D receiver operating characteristic analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8093–8115, Nov. 2020.

[47] J. Kerekes, "Receiver operating characteristic curve confidence intervals and regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 251–255, Apr. 2008.

[48] C. Ferri, J. Hernández-Orallo, and P. Flach, "A coherent interpretation of AUC as a measure of aggregated classification performance," in *Proc. Int. Conf. Mach. Learn.*, 2011, pp. 657–664.

[49] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[50] Q. Wang, Y. Tang, and P. M. Atkinson, "The effect of the point spread function on downscaling continua," *ISPRS J. Photogramm. Remote Sens.*, vol. 168, pp. 251–267, Oct. 2020.

[51] X. Wang, K. Tan, Q. Du, Y. Chen, and P. Du, "CVA2E: A conditional variational autoencoder with an adversarial training process for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5676–5692, Aug. 2020.

[52] X. Wang, K. Tan, Q. Du, Y. Chen, and P. Du, "Caps-TripleGAN: GAN-assisted capsnet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7232–7245, May 2019.

[53] F. Luo, H. Huang, Z. Ma, and J. Liu, "Semisupervised sparse manifold discriminative analysis for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6197–6211, Oct. 2016.

[54] F. Luo, Z. Zou, J. Liu, and Z. Lin, "Dimensionality reduction and classification of hyperspectral image via multistructure unified discriminative embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2021.3128764.

[55] C. Kwan, D. Gribben, B. Ayhan, S. Bernabe, A. Plaza, and M. Selva, "Improving land cover classification using extended multi-attribute profiles (EMAP) enhanced color, near infrared, and LiDAR data," *Remote Sens.*, vol. 12, no. 9, p. 1392, Apr. 2020.

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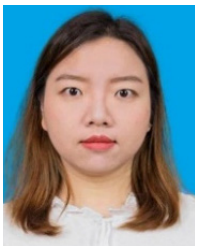


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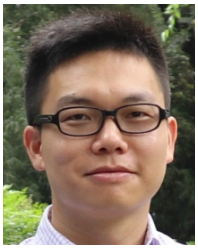
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